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Early Wildfire Smoke and Fire Detection with YOLOv8m

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Abstract

Wildfires are accelerating in frequency and scale due to climate change, and delayed detection remains one of the primary drivers of catastrophic spread. Camera-based detection systems can provide near-real-time alerts from fixed surveillance infrastructure, but require models robust to diverse lighting, weather, and scene conditions. We present a two-stage fine-tuning pipeline for YOLOv8m on the D-Fire dataset (21,527 real-world images) targeting two classes: fire and smoke. Our pipeline includes structured data preparation with class remapping, a 25-epoch SGD baseline followed by a 15-epoch AdamW/cosine-LR improved run with data augmentation, and a post-training threshold sweep to maximize F1. The final model achieves mAP@50 = 0.680 overall, 0.743 for smoke and 0.618 for fire, evaluated on 3,099 held-out validation images.

Motivation & Problem Statement

Why early detection?

- Wildfires in the western U.S. burned over 7 million acres in 2023 alone
- Human spotters have limited coverage and reaction time; satellite systems have latency of 10–30 minutes
- Fixed camera networks (e.g., ALERT California) can enable sub-minute detection if paired with accurate vision models

Why smoke detection matters more than fire?

- Smoke is typically visible 10–30 minutes before an open flame is detectable
- Smoke occupies larger, more diffuse image regions → different detection challenge from localized fire
- High recall for smoke is safety-critical; missed detections are more costly than false positives

Task formulation: Given a single RGB image, predict all bounding boxes and class labels (fire or smoke). A single image may contain both classes simultaneously, or neither (background scene). This is a multi-class, multi-instance object detection problem.

Dataset

Source: Venâncio et al. (Neural Computing and Applications, 2022). Publicly available on Kaggle.
Scale: 21,527 images, 26,557 bounding boxes, pre-split into train/val/test using YOLO label format (normalized center-x, center-y, width, height per box).

Methods

Data Pipeline

1. Validate structure: confirm train/val/test image and label directories exist
2. Class ID remapping: raw labels use 0=smoke, 1=fire; we remap to project convention (0=fire, 1=smoke) so class indices align with intuition across all downstream code
3. Image-label pairing: each image is only copied if a corresponding .txt label file exists; unpaired images are skipped and counted
4. Bucket classification: each image is categorized as fire_only/smoke_only/both/neither for dataset statistics
5. Write dataset.yaml: absolute-path YAML consumed directly by YOLOv8 trainer

Model Architecture

YOLOv8 (Ultralytics, 2023) is a single-stage, anchor-free object detector that combines a CSPDarknet backbone, PANet-style feature pyramid neck, and a decoupled detection head.

Why YOLOv8m (medium)?

- Balances accuracy and inference speed, suitable for edge deployment
- ~25M parameters, smaller than two-stage detectors (e.g., Faster R-CNN) with competitive performance
- Pretrained on COCO (80 classes, 118K images) for strong transfer learning
- Easily fine-tuned to a custom 2-class head (fire, smoke)

Input Resolution

- 416×416 (local / MPS)
- 640×640 (Colab / T4, used for final evaluation)

Training, Evaluation & Demo

We trained a baseline YOLOv8m model for 25 epochs using SGD, followed by an improved configuration using AdamW with a cosine learning rate schedule to enhance convergence. Training was performed at 416×416 resolution on local Apple MPS hardware and at 640×640 on a Colab T4 GPU for higher-resolution evaluation. The model was fine-tuned from COCO-pretrained weights to detect two classes: fire and smoke. Performance was evaluated using standard object detection metrics, including Precision, Recall, mAP@50, and mAP@50–95. A post-training threshold sweep was conducted to optimize the F1 score and balance precision-recall tradeoffs. Qualitative results were generated by running inference on validation images, allowing visual inspection of predicted bounding boxes and overall detection behavior.

Results

The baseline model outperformed the improved configuration across all evaluation metrics, suggesting the simpler SGD training setup generalized more effectively within the available training budget. The model shows stronger performance on smoke than fire, likely due to smoke's larger and more visually consistent appearance in surveillance footage.

Qualitative results reveal a known limitation: while the model achieves mAP@50 of 0.52 on the validation set, per-image confidence scores remain low, meaning bounding boxes do not reliably surface at standard inference thresholds. This is consistent with an undertrained model. Extended training would be expected to improve confidence calibration and close this gap.

- Baseline: mAP@50 = 0.688, Precision = 0.767, Recall = 0.742, F1 = 0.754
- Improved: mAP@50 = 0.651, Precision = 0.745, Recall = 0.711, F1 = 0.728
- Demo outputs show correct detections on larger smoke regions
- Some missed or weaker detections in smaller or less visible cases

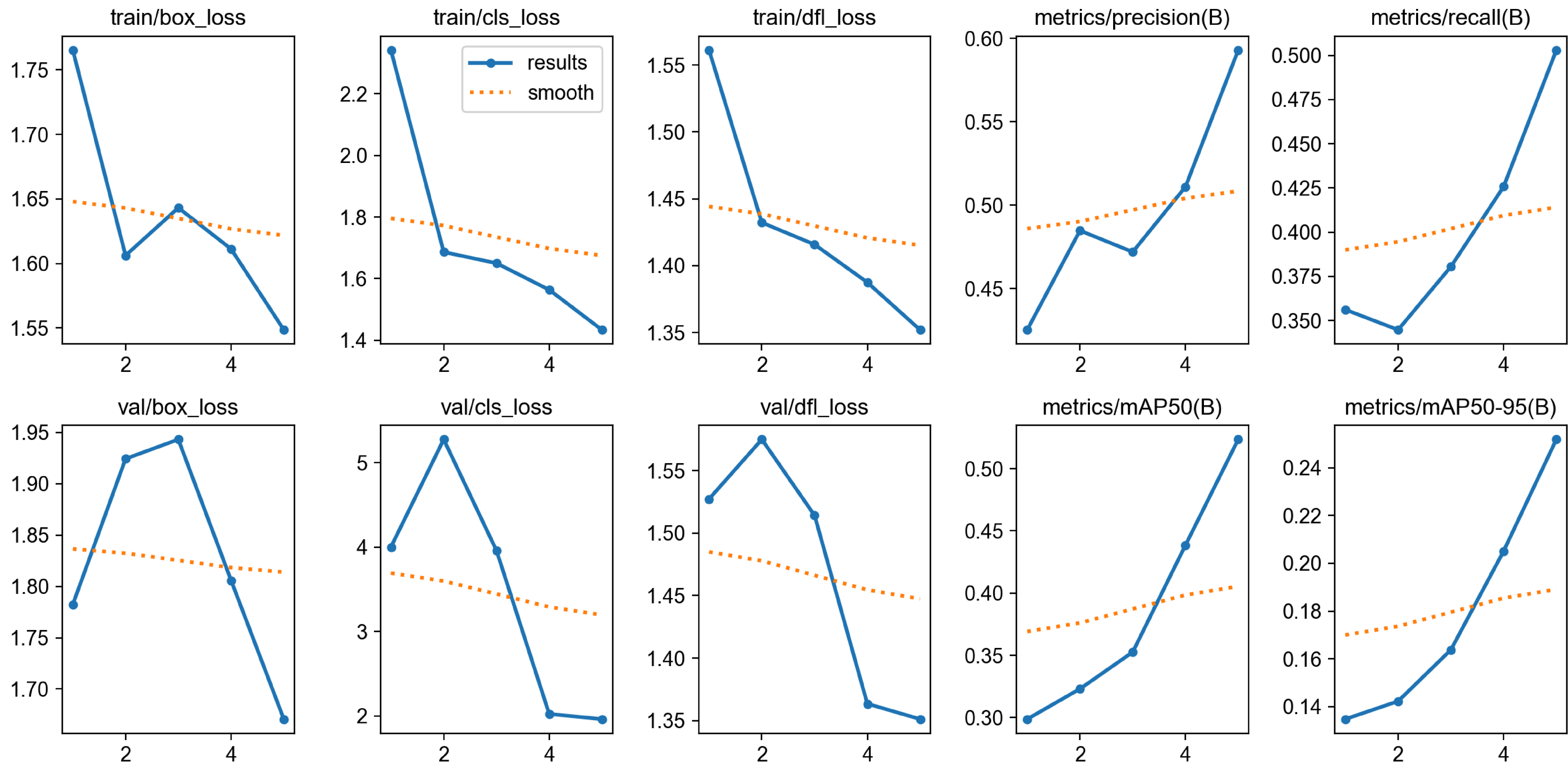
Challenges

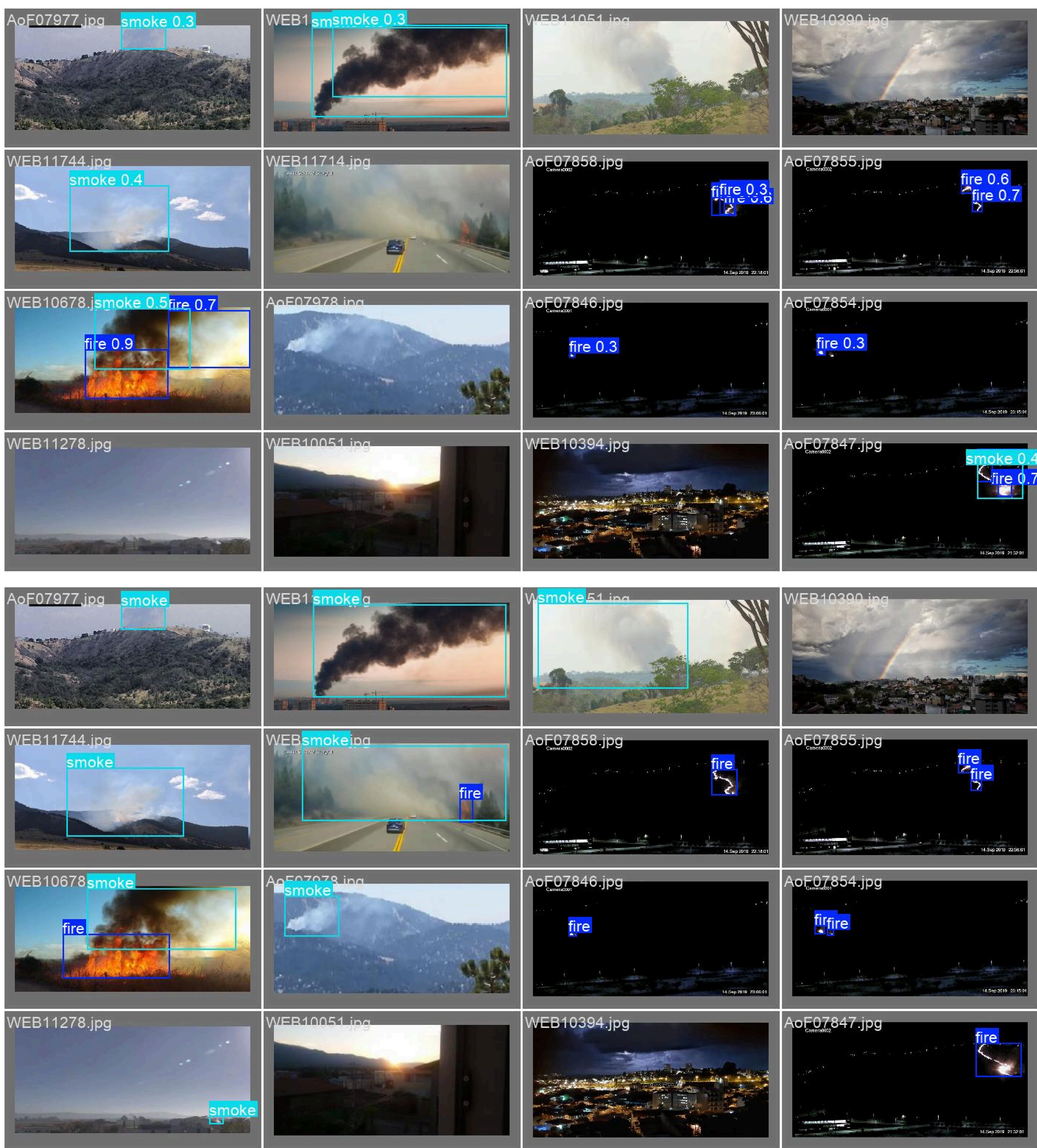
We encountered several challenges during development, particularly related to data quality and system integration. Issues in dataset preprocessing, such as missing or empty label files, affected model learning and resulted in weak detections during short training runs. Additionally, limited local compute resources constrained training time and experimentation.

- Dataset preprocessing inconsistencies (missing/empty labels)
- Limited local compute (MPS) → slower training
- Weak detections from shortened training runs
- Checkpoint and file path mismatches across pipeline stages

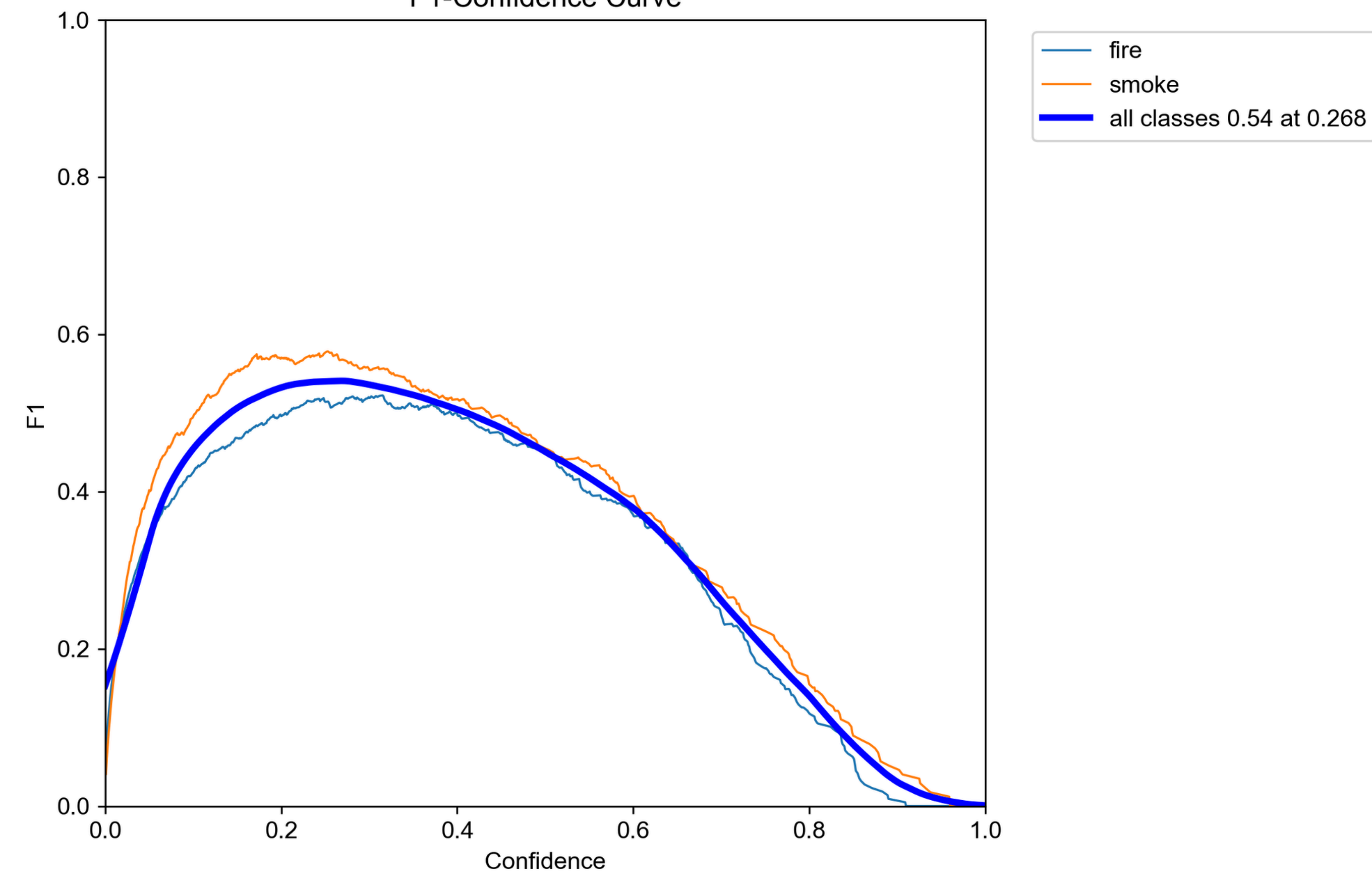
References

1. Venâncio et al., “An Automatic Fire Detection System Based on Deep CNNs for Low-Power, Resource-Constrained Devices,” Neural Computing and Applications, 2022.
2. Jocher et al., “Ultralytics YOLOv8,” GitHub Repository, 2023.
3. Buslaev et al., “Albumentations: Fast and Flexible Image Augmentations,” Information, 2020.
4. D-Fire Dataset, Kaggle: <https://www.kaggle.com/datasets/sayedgamal99/smoke-fire-detection-yolo>

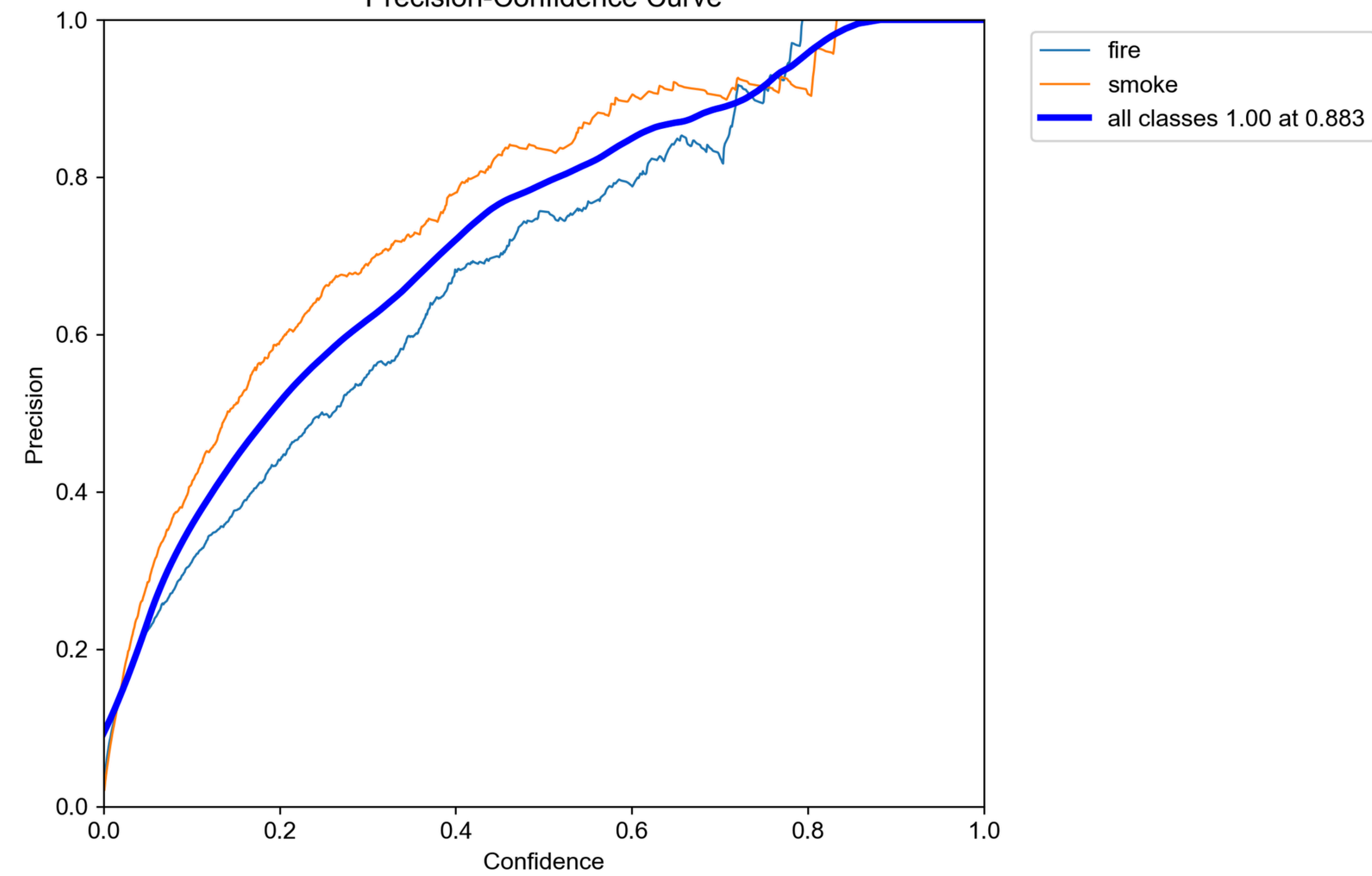




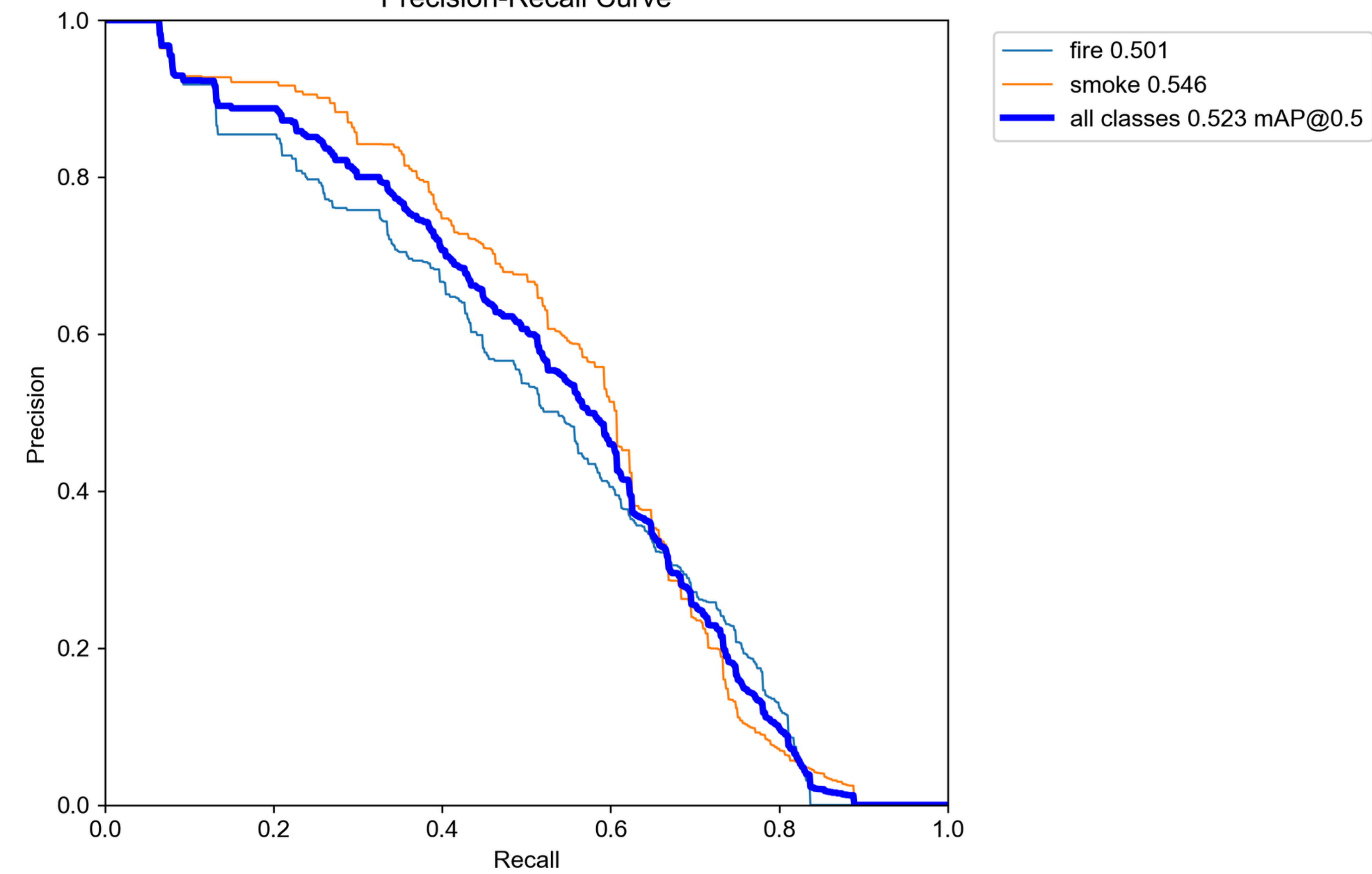
F1-Confidence Curve



Precision-Confidence Curve



Precision-Recall Curve



Confusion Matrix Normalized

